Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

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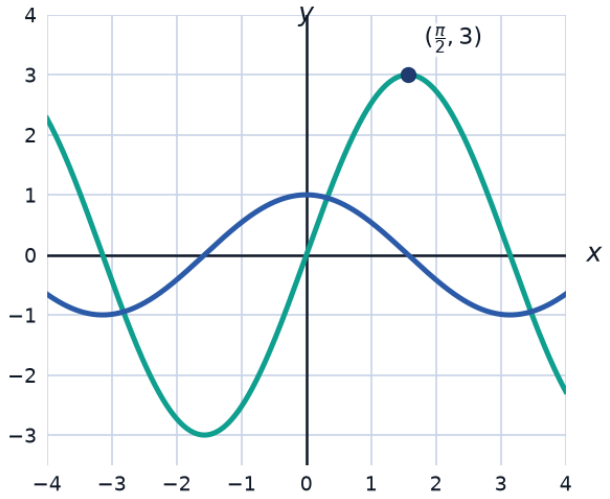
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Reciprocal Trigonometric Functions

Express $f(x) = \frac{3x^2 + 2x - 11}{(x + 2)}$ in the form $mx + c + \frac{r}{(x + 2)}$, and state the value of r.

2 [Foundational] • Target: Compound Angle Addition Formulas

Express $f(x) = \frac{3x^2 + 5x - 16}{(x + 3)}$ in the form $mx + c + \frac{r}{(x + 3)}$, and state the value of r.



3 [Foundational] • Target: Double Angle Formulas

Express $f(x) = \frac{x^2 + 3x - 7}{(x - 1)}$ in the form $mx + c + \frac{r}{(x - 1)}$, and state the value of r.

4 [Foundational] • Target: Linear Trigonometric Combinations (R-form)

Express $f(x) = \frac{2x^2 - 6x + 9}{(x - 2)}$ in the form $mx + c + \frac{r}{(x - 2)}$, and state the value of r.

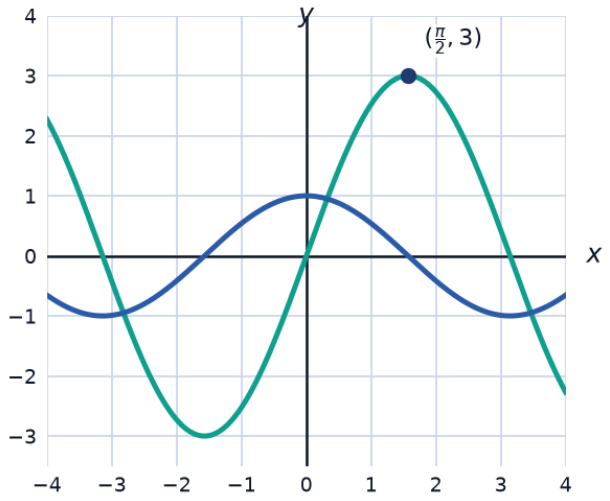
5 [Foundational] • Target: Reciprocal Trigonometric Functions

Express $f(x) = \frac{x^2 + x - 11}{(x + 3)}$ in the form $mx + c + \frac{r}{(x + 3)}$, and state the value of r.

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6 [Foundational] • Target: Compound Angle Addition Formulas

Express $f(x) = \frac{x^2 - 3x - 4}{(x)}$ in the form $mx + c + \frac{r}{(x)}$, and state the value of r.



7 [Foundational] • Target: Double Angle Formulas

Express $f(x) = \frac{3x^2 - 4x + 2}{(x)}$ in the form $mx + c + \frac{r}{(x)}$, and state the value of r.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

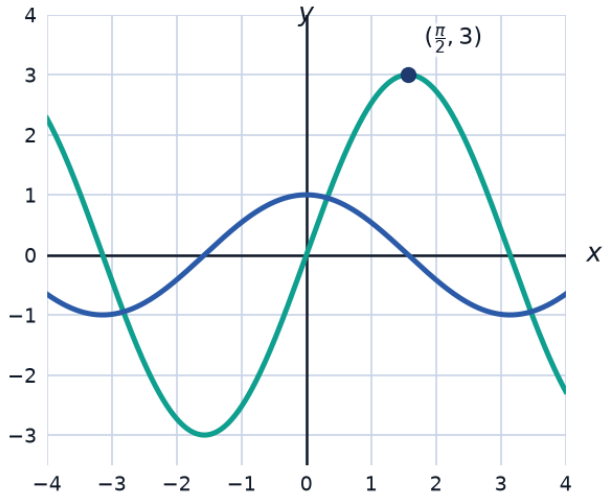
Express $f(x) = \frac{3x^2 + 4x + 2}{(x)}$ in the form $mx + c + \frac{r}{(x)}$, and state the value of r.

9 [Intermediate] • Target: Reciprocal Trigonometric Functions

Express $f(x) = \frac{3x^2 - 10x + 9}{(x - 2)}$ in the form $mx + c + \frac{r}{(x - 2)}$, and state the value of r.

10 [Intermediate] • Target: Compound Angle Addition Formulas

Express $f(x) = \frac{x^2 + 5x + 8}{(x + 4)}$ in the form $mx + c + \frac{r}{(x + 4)}$, and state the value of r.



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11 [Intermediate] • Target: Double Angle Formulas

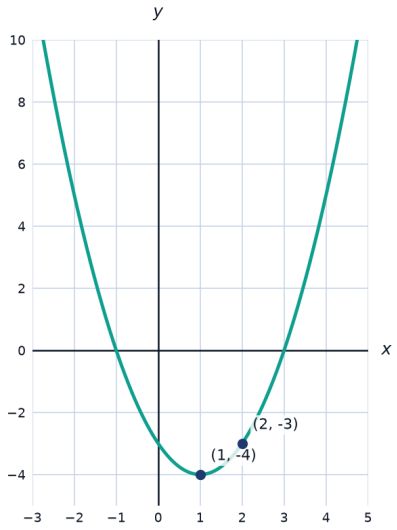
Express $f(x) = \frac{3x^2 - 9x + 4}{(x - 3)}$ in the form $mx + c + \frac{r}{(x - 3)}$, and state the value of r.

12 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Express $f(x) = \frac{x^2 - 3x + 5}{(x - 3)}$ in the form $mx + c + \frac{r}{(x - 3)}$, and state the value of r.

13 [Intermediate] • Target: Reciprocal Trigonometric Functions

The diagram shows the graph of the function $y = f(x)$. The function is of the form $f(x) = x^2 + bx + c$. The graph passes through the point $(2, -3)$ and has a minimum value of -4 at $x = 1$. Find the value of b.



14 [Intermediate] • Target: Compound Angle Addition Formulas

Solve $|x + 1| = |2x - 1|$. Give the largest solution.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Solve $|x + 2| = |2x - 3|$. Give the largest solution.

16 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Solve $|2x + 1| = |x - 1|$. Give the largest solution.



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17 [Challenging] • Target: Reciprocal Trigonometric Functions • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Solve $|x - 5| = |2x + 5|$. Give the largest solution.

18 [Challenging] • Target: Compound Angle Addition Formulas • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Solve $|3x - 6| = |-2x|$. Give the largest solution.

19 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 4.02 The Poisson Distribution

Find $f^{-1}(x)$ if $f(x) = \frac{6}{x}$.

20 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 5.01 Further Integration Techniques

Find $f^{-1}(x)$ if $f(x) = \frac{3}{x}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Reciprocal Trigonometric Functions

Simplify the trigonometric expression $\frac{\sec x - \cos x}{\tan x}$.

a. $\sin x$

b. $\cos x$

c. $\cot x$

d. $\csc x$

22 [Foundational] • Target: Reciprocal Trigonometric Functions

Find the exact value of $\csc\left(\frac{5\pi}{6}\right)$.

a. 2

b. -2

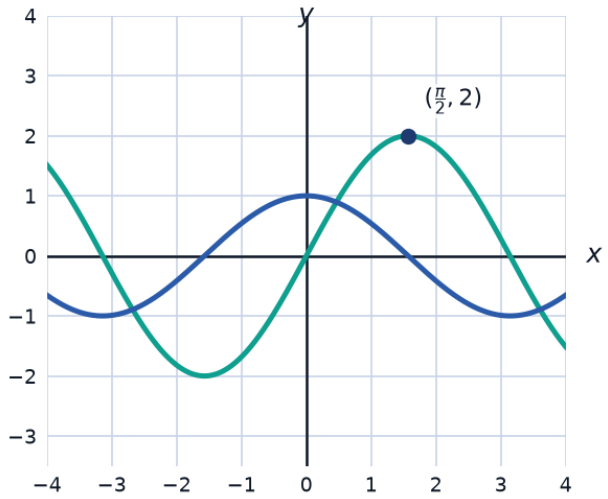
c. $\frac{2}{\sqrt{3}}$

d. $-\frac{2}{\sqrt{3}}$

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23 [Foundational] • Target: Reciprocal Trigonometric Functions

Which identity relates $\cot\theta$ and $\csc\theta$ for all defined angles?



a. $1+\cot^2\theta=\csc^2\theta$

b. $1+\csc^2\theta=\cot^2\theta$

c. $\cot^2\theta-\csc^2\theta=1$

d. $\csc\theta=1+\cot\theta$

24 [Foundational] • Target: Compound Angle Addition Formulas

Use compound angle identities to find the exact value of $\cos(75^\circ)=\cos(45^\circ+30^\circ)$.

a. $\frac{\sqrt{6}-\sqrt{2}}{4}$

b. $\frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{2\sqrt{2}}$

d. $\frac{\sqrt{2}-\sqrt{6}}{4}$

25 [Foundational] • Target: Compound Angle Addition Formulas

Simplify $\sin(x+y)\cos y-\cos(x+y)\sin y$.

a. $\sin x$

b. $\cos x$

c. $\sin(x+2y)$

d. $\cos(x-y)$

26 [Foundational] • Target: Compound Angle Addition Formulas

If $\tan A=2$ and $\tan B=3$, what is $\tan(A+B)$?

a. -1

b. 1

c. 5

d. $\frac{5}{7}$

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27 [Foundational] • Target: Double Angle Formulas

If $\sin\theta = \frac{3}{5}$ and θ is acute, what is $\cos(2\theta)$?

a. $\frac{7}{25}$

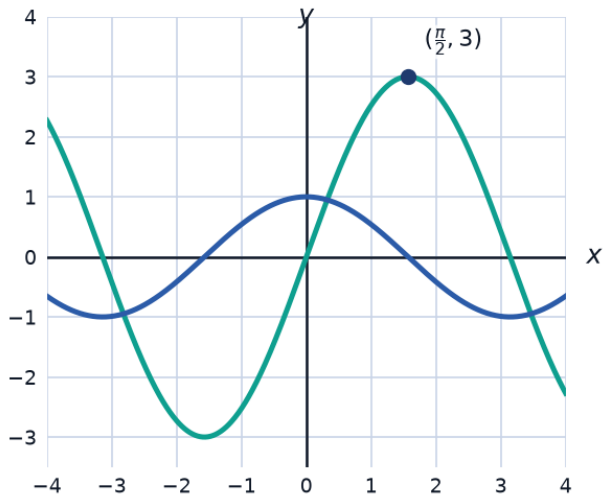
b. $\frac{24}{25}$

c. $-\frac{7}{25}$

d. $\frac{16}{25}$

28 [Foundational] • Target: Double Angle Formulas

Under the same conditions ($\sin\theta = \frac{3}{5}$ acute), what is $\sin(2\theta)$?



a. $\frac{24}{25}$

b. $\frac{7}{25}$

c. $\frac{12}{25}$

d. $\frac{6}{5}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double Angle Formulas

Express $\cos^2 x$ in terms of a first-power cosine function.

a. $\frac{1+\cos(2x)}{2}$

b. $\frac{1-\cos(2x)}{2}$

c. $\frac{1+\sin(2x)}{2}$

d. $1-\sin^2 x$

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30 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Express $3\sin x + 4\cos x$ in the form $R\sin(x+\alpha)$ with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

a. $5\sin(x + 0.927)$

b. $5\sin(x + 0.643)$

c. $7\sin(x + 0.927)$

d. $25\sin(x + 0.927)$

31 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

What is the maximum value of the function $f(x) = 5\cos x - 12\sin x + 3$?

a. 16

b. 13

c. 10

d. 20

32 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Solve $\sin(2x) = \cos x$ for $x \in [0, 2\pi)$.

a. $x = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $x = \frac{\pi}{6}, \frac{5\pi}{6}$

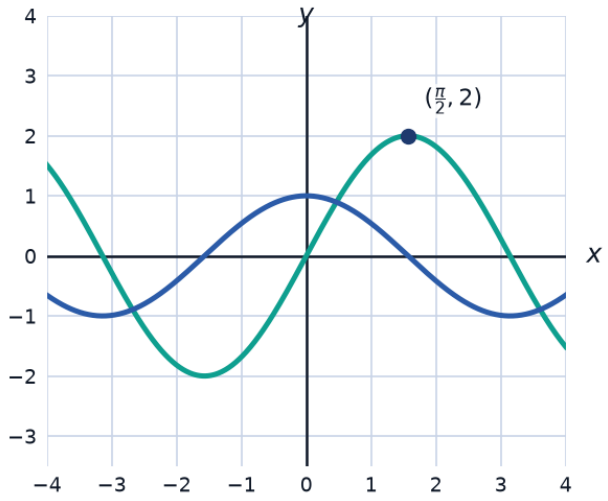
c. $x = 0, \pi$

d. $x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

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33 [Intermediate] • Target: Reciprocal Trigonometric Functions

Find $f^{-1}(x)$ if $f(x) = \frac{4}{x}$.



a. $\frac{4}{x}$

b. $-\frac{4}{x}$

c. $\frac{1}{4x}$

d. $\frac{x}{4}$

34 [Intermediate] • Target: Compound Angle Addition Formulas

Find $f^{-1}(x)$ if $f(x) = \frac{5}{x}$.

a. $\frac{1}{5x}$

b. $\frac{x}{5}$

c. $-\frac{5}{x}$

d. $\frac{5}{x}$

35 [Intermediate] • Target: Double Angle Formulas

Find $f^{-1}(x)$ if $f(x) = \frac{2}{x}$.

a. $\frac{2}{x}$

b. $\frac{1}{2x}$

c. $\frac{x}{2}$

d. $-\frac{2}{x}$

36 [Intermediate] • Target: Linear Trigonometric Combinations (R-form)

Solve $|3x| = |-x + 4|$. Give the largest solution.

a. 2

b. 0

c. -1

d. 1

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37 **[Intermediate]** • **Target: Reciprocal Trigonometric Functions**

Find the domain of $f(x) = \sin^{-1}(4x + 2)$.

a. $-\frac{3}{4} \leq x \leq -\frac{1}{4}$

b. $-\frac{3}{4} < x < -\frac{1}{4}$

c. $-\frac{1}{4} \leq x \leq -\frac{3}{4}$

d. $-1 \leq x \leq 1$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Compound Angle Addition Formulas** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Find the domain of $f(x) = \sin^{-1}(6x + 2)$.

a. $-\frac{1}{2} < x < -\frac{1}{6}$

b. $-\frac{1}{2} \leq x \leq -\frac{1}{6}$

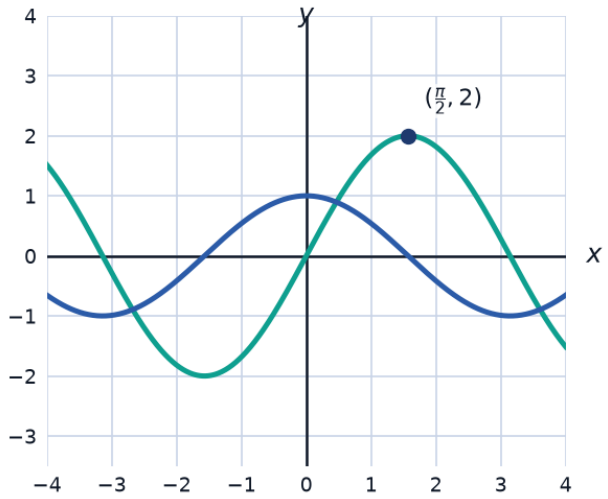
c. $-\frac{1}{6} \leq x \leq -\frac{1}{2}$

d. $-1 \leq x \leq 1$

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39 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 4.02 The Poisson Distribution

Find the domain of $f(x) = \sin^{-1}(3x + 3)$.



a. $-\frac{4}{3} \leq x \leq -\frac{2}{3}$

b. $-\frac{4}{3} < x < -\frac{2}{3}$

c. $-\frac{2}{3} \leq x \leq -\frac{4}{3}$

d. $-1 \leq x \leq 1$

40 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 5.01 Further Integration Techniques

Find the domain of $f(x) = \sin^{-1}(6x - 2)$.

a. $\frac{1}{6} < x < \frac{1}{2}$

b. $-1 \leq x \leq 1$

c. $\frac{1}{2} \leq x \leq \frac{1}{6}$

d. $\frac{1}{6} \leq x \leq \frac{1}{2}$

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41 [Challenging] • Target: Reciprocal Trigonometric Functions • Cross-Lesson: 5.02 Differential Equations

Find the domain of $f(x) = \sin^{-1}(6x - 3)$.

a. $-1 \leq x \leq 1$

b. $\frac{1}{3} < x < \frac{2}{3}$

c. $\frac{2}{3} \leq x \leq \frac{1}{3}$

d. $\frac{1}{3} \leq x \leq \frac{2}{3}$

42 [Challenging] • Target: Compound Angle Addition Formulas • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Find the domain of $f(x) = \sin^{-1}(4x - 4)$.

a. $\frac{5}{4} \leq x \leq \frac{3}{4}$

b. $\frac{3}{4} < x < \frac{5}{4}$

c. $-1 \leq x \leq 1$

d. $\frac{3}{4} \leq x \leq \frac{5}{4}$

43 [Challenging] • Target: Double Angle Formulas • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Find the angle between $\mathbf{u} = (2; 6; 9)$ and $\mathbf{v} = (6; -2; 0)$.

a. 90°

b. 104°

c. 72°

d. 69°

44 [Challenging] • Target: Linear Trigonometric Combinations (R-form) • Cross-Lesson: 6.01 Matrices (AI HL)

Find the angle between $\mathbf{u} = (6; 2; 3)$ and $\mathbf{v} = (2; -6; 0)$.

a. 92°

b. 90°

c. 100°

d. 76°

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45 [Challenging] • Target: Reciprocal Trigonometric Functions • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

Find the angle between $\mathbf{u} = (3; 4; 12)$ and $\mathbf{v} = (4; -3; 0)$.

a. 92°

b. 77°

c. 102°

d. 90°

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